

AWM: Abductive World Modeling via Hierarchical State Evidence

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Abstract

World models are commonly judged by the accuracy of their future predictions. This criterion is necessary but incomplete: a representation may predict a consequence without exposing which evidence supports it or how the prediction should change when that evidence is altered. We propose abductive state modeling, a formulation that treats a world state as a latent interface which organizes partial visual evidence so that it can explain task-relevant factors, support prediction, and respond selectively to controlled interventions. We instantiate the formulation with an Abductive World Model (AWM) built on a frozen predictive video backbone. Its Hierarchical Abductive State Pyramid (HASP) progressively attributes patch evidence to three complementary state factors—Entity, Dynamic, and Relation—corresponding to what exists, how it changes, and which interactions can alter the future. These factors are exposed through one task-independent state interface and consumed by lightweight task readouts. We evaluate the interface with factor-level probes, matched-capacity prediction, state and input interventions, and transfer across physical reasoning, event reasoning, and human-action understanding. The resulting protocol makes explainability, predictability, and intervention consistency measurable properties of a world state, without assuming that the learned factors are uniquely identifiable causal variables.

1 Introduction

World models aim to compress observations into an internal state from which an agent can anticipate consequences, reason about alternatives, and reuse knowledge in new situations. For visual interaction, such a state cannot be only a global appearance summary. It should preserve the entities that are present, the way their states evolve, and the interactions through which one entity can change another. Predictive video models have shown that learning in a latent space can be substantially more efficient than reconstructing pixels (Assran et al., 2023; Bardes et al., 2024; Assran et al., 2025). Yet latent prediction alone does not specify how visual evidence is organized inside the state or whether the resulting representation is useful beyond the readout that was trained with it.

This gap is important for two reasons. First, the same video can support different queries: a physical predictor may need contact and time-to-contact, an event reasoner may need object identity and temporal order, and an action recognizer may need both object appearance and motion. A representation that is useful for one query may still be an entangled collection of shortcuts. Second, a high probe score only shows that information is available. It does not show that the prediction pathway uses the information, that the information is attached to the intended unit, or that the representation remains useful when the evidence is changed.

We address this problem by asking a more precise question: what empirical properties should a latent representation satisfy before it is called a reusable world state? We formulate three requirements. A state must be explainable: the factors it claims to organize must be decodable at the corresponding granularity. It must be predictive: a controlled-capacity readout from the state must support future or task-level prediction. Finally, it must be intervention-consistent: when evidence associated with one factor is masked, swapped, or

rewired, the corresponding state and prediction should change more than unrelated outputs. These requirements turn interpretability into a falsifiable evaluation protocol rather than a qualitative visualization.

The term abductive describes the direction of inference behind this protocol. Given incomplete observations and a possible consequence, the model organizes the available evidence into a state that could explain what is observed and make the consequence predictable. This is deliberately weaker than causal discovery. Our current deterministic state extractor does not identify ground-truth causal variables or produce a calibrated posterior over hypotheses. Instead, it makes explanatory factor hypotheses explicit and tests them by combining probes, matched readouts, and controlled interventions.

We instantiate the formulation with AWM, which reuses a frozen V-JEPA 2 encoder and predictor and adds HASP, a hierarchical state interface. The hierarchy is induced by a conditional dependency: first establish which entities are present, then encode how those entities change, and finally represent which pairs of entities have a configuration capable of changing the future. The three stages are therefore called the Entity, Dynamic, and Relation Attributors. They are not independent task heads. Each stage refines the evidence organized by the preceding stage and retains a residual path to lower-level evidence. A task-specific readout may use the resulting state, but the state interface itself is independent of the task label space.

The central claim of this paper is methodological: a world model should be evaluated by the joint behavior of its evidence, state, and consequence. We make the following contributions:

- We formulate abductive state modeling, defining a reusable world state through explainability, predictive sufficiency, and intervention consistency.
- We introduce HASP, a residual hierarchical interface that organizes patch evidence into Entity, Dynamic, and Relation states while preserving the strength of a pre-trained predictive backbone.
- We give an analysis-first evaluation protocol that separates information presence from functional use, and functional use from compositional reuse.
- We provide a unified experimental interface for physical prediction, event reasoning, and action understanding, with explicit boundaries on what probe and intervention results can establish.

2 Related Work

Latent predictive world models. JEPA-style methods predict representations rather than pixels, providing an efficient route to visual prediction (Assran et al., 2023; Bardes et al., 2024; Assran et al., 2025). Dreamer and related latent-dynamics models learn compact states for imagination and control (Hafner et al., 2020; 2023). AWM is complementary to these approaches: it reuses a predictive backbone but treats the intermediate state interface as a measurable scientific object. The question is not only whether a future latent is accurate, but also which factors are organized in the state and how the prediction responds to evidence interventions.

Object-centric and relational representations. Slot Attention and its video extensions organize visual features into object-centered entities (Locatello et al., 2020; Kipf et al., 2022; Seitzer et al., 2023; Wu et al., 2022). Interaction Networks, Neural Relational Inference, and related models represent pairwise interactions or physical dynamics (Battaglia et al., 2016; Kipf et al., 2018; Lin et al., 2020). HASP combines these ideas in one conditional residual path and, importantly, does not assume that architectural names imply semantic specialization. Specialization is an empirical hypothesis tested at the native slot, temporal, and pair granularities.

Causal, invariant, and explanatory representations. Causal representation learning and independent-mechanism approaches emphasize stability under distribution

changes (Scholkopf et al., 2021; Parascandolo et al., 2018; Arjovsky et al., 2019; Khemakhem et al., 2020). Abductive learning connects observations and consequences to explanatory latent causes (Peirce, 1903; Dharami et al., 2021; Evans et al., 2021). Our intervention protocol is inspired by these goals, but the scope is explicit: current results establish functional and selective dependence, not identification of true causal variables.

Structured video evaluation. Controlled physical and event benchmarks make object, motion, contact, and temporal factors measurable, while unconstrained egocentric video tests whether a state interface transfers to semantic action labels (Yi et al., 2020; Bear et al., 2021; Damen et al., 2022). We use these task spaces as complementary tests of one state interface rather than as separate architectural contributions.

3 Problem Formulation

Let $X_{1:t}$ denote the observed video context and let $Y_{t+\Delta}$ denote a future outcome, event query, or action label. We seek a latent state $S_t = (E_t, D_t, R_t)$, where E_t describes entities, D_t describes their temporal evolution, and R_t describes pairwise interactions. A useful state should mediate the relationship between evidence and consequence:

$$p(Y | X) = \sum_{E,D,R} p(Y | E, D, R)p(R | E, D)p(D | E, X)p(E | X). \quad (1)$$

Equation 1 is a structural lens rather than a claim that the current deterministic Attributors estimate calibrated posteriors.

The three requirements can be written as an evaluation criterion. For a factor $c_k \in \{E, D, R\}$, let q_k be a matched-capacity probe and let $\mathcal{I}_k$ be an intervention that changes evidence associated with c_k . We require: (i) $q_k(S_k)$ predicts the native factor, (ii) a readout $r(S_t)$ supports Y , and (iii) the target change under $\mathcal{I}_k$ exceeds the change in matched non-target outputs. A compact conceptual objective is

$$\mathcal{L}_{\text{state}} = \mathcal{L}_{\text{explain}} + \mathcal{L}_{\text{predict}} + \lambda \mathcal{L}_{\text{intervene}} + \beta \mathcal{R}_{\text{complexity}}. \quad (2)$$

In this work the objective is evaluated retrospectively: the backbone and HASP are frozen, and independent readouts and interventions provide evidence for each term.

The abductive perspective is useful when several explanations are compatible with a partial observation. For a candidate $h = (E, D, R)$ and an observed consequence Y , we define the conceptual score

$$s(h; X, Y) = \log p(X | h) + \log p(Y | h) - \lambda \Omega(h), \quad h^* = \arg \max_h s(h; X, Y). \quad (3)$$

The present paper validates the ingredients of this score through decoding and intervention experiments. End-to-end candidate generation, posterior calibration, and explicit hypothesis ranking are left as a controlled extension rather than claimed as completed capabilities.

4 Method

4.1 State interface for predictive world modeling

Rather than defining a world model by a particular task head, we define its output as a structured state interface. Given partial visual evidence, the interface should expose three complementary factors, support a downstream consequence, and provide units that can be selectively perturbed:

$$S_t = (S_t^E, S_t^D, S_t^R). \quad (4)$$

Entity describes what is present, Dynamic describes how entities evolve, and Relation describes which pairs may alter one another’s future. A factor is functionally meaningful when its native units are readable, useful for prediction, and more sensitive to matched target interventions than to non-target interventions. This is an empirical criterion, not an assumption that the learned variables are uniquely causal.

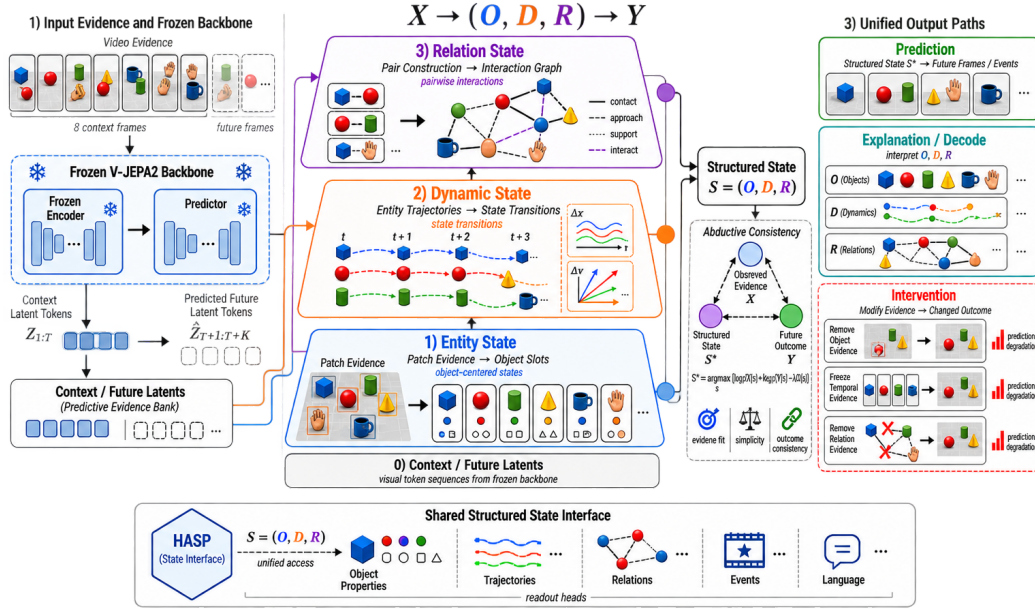


Figure 1: AWM organizes visual evidence into a reusable state interface. The encoder and predictor are frozen in the analysis protocol; HASP progressively attributes evidence to Entity, Dynamic, and Relation states, and a lightweight readout maps the shared interface to a task-specific output. The predictor supplies a future hypothesis from context only.

4.2 Predictive visual backbone

Let $X_{1:t}$ denote the observed video context. We use a pretrained latent-video predictive backbone, represented by a visual encoder E and a predictor P ,

$$Z_c = E(X_{1:t}), \quad \hat{Z}_f = P(Z_c), \quad Z_c, \hat{Z}_f \in \mathbb{R}^{B \times T \times N \times d}. \quad (5)$$

The encoder converts the observation into spatiotemporal patch evidence Z_c , while the predictor produces a future hypothesis $\hat{Z}_f$ from context alone. The backbone is chosen because latent prediction already provides rich evidence about appearance, motion, and temporal context; AWM therefore studies how such evidence is organized into a reusable state rather than relearning a larger visual feature extractor. In all inference protocols, ground-truth future frames are withheld, so $\hat{Z}_f$ is a prediction-conditioned evidence stream and not a second observation or a pixel reconstruction. The role of AWM is therefore to organize predictive evidence rather than relearn the backbone’s low-level visual representation.

4.3 HASP: hierarchical evidence attribution

Design principles. HASP is a conditional attribution path, not a collection of independent task heads. Its native units become progressively structured: Entity uses slots, Dynamic uses slot–time states, and Relation uses pair–time states. This progression follows the dependencies needed to explain a scene: interactions are defined over entities and their evolution is defined over time. Residual routing preserves lower-level evidence, so later abstraction does not become a destructive bottleneck and each level remains directly intervenable. We define

$$S_t = (S_t^E, S_t^D, S_t^R), \quad (6)$$

where Entity describes what is present, Dynamic describes how entities evolve, and Relation describes which entity pairs can affect one another’s future. These labels are architectural hypotheses, not assumptions that the learned variables are uniquely causal. Their validity is tested by native factor decoding, matched-capacity task readouts, and targeted interventions.

Conditional organization. HASP imposes the following conditional organization of evidence:

$$p(S^E, S^D, S^R \mid Z_c, \hat{Z}_f) \propto p(S^E \mid Z_c) p(S^D \mid S^E, Z_c, \hat{Z}_f) p(S^R \mid S^E, S^D, Z_c, \hat{Z}_f), \quad (7)$$

where the factorization is a structural inductive bias rather than a claim that the deterministic Attributors estimate calibrated posteriors. In implementation, the corresponding deterministic path is

$$S^E = F_E(Z_c), \quad (8)$$

$$S^D = F_D(S^E, Z_c, \hat{Z}_f) + G_D(S^E, Z_c), \quad (9)$$

$$S^R = F_R(S^E, S^D, Z_c, \hat{Z}_f) + G_R(S^D, Z_c), \quad (10)$$

$$S = \text{ReadoutInterface}(S^E, S^D, S^R). \quad (11)$$

Here F_l attributes new evidence at level l , whereas G_l carries forward lower-level evidence. The residual paths do not guarantee disentanglement; they make information preservation and slot-, slot-time-, and pair-time-level interventions explicit.

Forward computation. The complete inference path is

$$X_{1:t} \xrightarrow{E} Z_c \xrightarrow{P} \hat{Z}_f \xrightarrow{F_E, F_D, F_R} (S^E, S^D, S^R) \xrightarrow{R_\tau} \hat{Y}_\tau. \quad (12)$$

4.4 Entity Attributor: object-centered evidence

The Entity Attributor changes the unit from an unordered patch set to a fixed collection of object-centered slots, $S^E \in \mathbb{R}^{B \times K \times d}$. Learned queries provide the slot interface and are iteratively refined by normalized cross-attention:

$$Q_E^{(0)} = \text{LearnedQueries}, \quad (13)$$

$$A_E^{(l)} = \text{CrossAttn}(\text{LN}(Q_E^{(l)}), \text{LN}(Z_c)), \quad (14)$$

$$Q_E^{(l+1)} = Q_E^{(l)} + \text{FFN}(A_E^{(l)}). \quad (15)$$

Cross-attention selects patch evidence relevant to each slot, while the residual update preserves an object-centered state across refinement. A slot is not assumed to be a guaranteed persistent track; permutation alignment and semantic interpretation are tested rather than encoded in its index.

Entity supervision is expressed at the same granularity,

$$\mathcal{L}_E = \lambda_{\text{pres}} \mathcal{L}_{\text{pres}} + \lambda_{\text{geom}} \mathcal{L}_{\text{geom}} + \lambda_{\text{state}} \mathcal{L}_{\text{state}}, \quad (16)$$

with presence, geometry, and object-state terms. These constraints turn object semantics from an informal interpretation into a measurable slot-level interface.

4.5 Dynamic Attributor: temporal transition evidence

Global temporal pooling can mix multiple entities and erase direction or event progress. The Dynamic Attributor therefore conditions on Entity identity and constructs a slot-time state. Its query construction is

$$Q_D^{(0)} = S^E + E_{\text{time}} + E_{\text{future}}, \quad (17)$$

where ϕ_E is a learned projection of the Entity slots, and the latter terms encode temporal position and future-hypothesis evidence. Context and predicted evidence are first associated with these queries and then processed temporally,

$$H_D = \text{CrossAttn}(Q_D^{(0)}, [Z_c; \hat{Z}_f]), \quad S^D = \text{TemporalTransformer}(H_D) + G_D(S^E, Z_c). \quad (18)$$

The output represents transitions of identified entities, not a second object detector; its native unit is slot-time, $S^D \in \mathbb{R}^{B \times K \times T \times d}$. It is intended to encode displacement, velocity,

direction, temporal order, and event progress while preserving the identity supplied by Entity.

The corresponding constraint is

$$\mathcal{L}_D = \lambda_{\text{traj}} \mathcal{L}_{\text{traj}} + \lambda_{\text{vel}} \mathcal{L}_{\text{vel}} + \lambda_{\text{order}} \mathcal{L}_{\text{order}}, \quad (19)$$

which anchors the representation to trajectories, motion variables, and temporal ordering instead of allowing static appearance shortcuts to explain all tasks.

4.6 Relation Attributor: pairwise interaction evidence

Let $\mathcal{P} = \{(i, j) : i < j\}$ denote candidate unordered entity pairs. The Relation Attributor changes the unit from an individual slot to an entity pair and constructs explicit common and relative evidence from Entity and Dynamic states,

$$r_{ij,t}^{(0)} = [S_i^E + S_j^E, |S_i^E - S_j^E|, S_{i,t}^D + S_{j,t}^D, |S_{i,t}^D - S_{j,t}^D|], \quad (20)$$

The sums encode shared state, the absolute differences encode relative state, and the Dynamic terms preserve when the relation changes. Pairwise projection followed by temporal aggregation gives

$$S_{ij,1:T}^R = \text{TemporalAttention}(\text{MLP}_R(r_{ij,1:T}^{(0)})) + G_R(S^D, Z_c). \quad (21)$$

The resulting pair-time state, $S^R \in \mathbb{R}^{B \times |\mathcal{P}| \times T \times d}$, can represent relative distance, contact, first-contact time, and time-to-contact. The combination of continuous geometry, event state, and event timing prevents Relation from collapsing into a purely binary contact classifier. Because the pair unit is explicit, contact masking, pair swapping, and relation rewiring are well-defined interventions.

Relation supervision is matched to this structure:

$$\mathcal{L}_R = \lambda_{\text{dist}} \mathcal{L}_{\text{dist}} + \lambda_{\text{contact}} \mathcal{L}_{\text{contact}} + \lambda_{\text{first}} \mathcal{L}_{\text{first}}. \quad (22)$$

The distance, contact, and first-contact terms constrain not only which objects are present but also when and how their interaction occurs.

4.7 Unified state interface and task readouts

The state extractor is shared while the readout is replaceable. The three structured outputs are exposed through the common interface S in Eq. 11. A task readout is a replaceable function

$$\hat{Y}_\tau = R_\tau(S^E, S^D, S^R), \quad (23)$$

whose label space is independent of the state extractor. A readout may use all levels or a selected subset, allowing physical prediction, event reasoning, and action recognition to test different combinations of the same state. For EK100, verb and noun heads are trained separately and their logits define a closed-set verb-noun action score; unseen combinations remain reportable as separate factors.

4.8 Training objective and analysis protocol

For a jointly trained state extractor, the conceptual objective is

$$\mathcal{L}_{\text{AWM}} = \mathcal{L}_E + \mathcal{L}_D + \mathcal{L}_R + \lambda_{\text{task}} \mathcal{L}_{\text{task}}. \quad (24)$$

The analysis protocol used here freezes the encoder, predictor, and HASP extractor and trains only native factor probes and task readouts. Input interventions alter visual evidence and rerun the frozen predictor and HASP; state interventions mask or replace slots, slot-time states, or pair states after HASP. Native probes test readability, task readouts test predictive sufficiency, input interventions test evidence propagation, state interventions test functional dependence, and cross-task readouts test reuse. These protocols establish functional organization without claiming perfect disentanglement, causal identification, or explicit posterior inference.

Table 1: Main comparison across datasets. All entries are percentage values reported without the percent sign. The best available value in each metric column is boldfaced. Here, “Bal. Acc.” and “Acc.” denote balanced accuracy and accuracy; all CLEVRER columns report option/question accuracy, and all EK100 entries are reported as Top-1/Top-5. AWM uses the shared HASP state interface, while V-JEPA2 is the corresponding backbone-only baseline.

Method	Physion++			CLEVRER		EK100 (T1/T5)		
	AUROC	Bal. Acc.	Acc.	Option Acc.	Question Acc.	Verb	Noun	Action
ORCA	59.15	55.94	56.64	—	—	—	—	—
V-JEPA 2	65.20	61.57	61.60	65.46	42.20	55.74/86.81	34.65/62.67	25.67/46.94
VideoMAE v2	67.61	62.60	62.86	—	—	59.54/82.57	46.37/68.79	35.96/54.57
AWM (Ours)	72.19	66.17	65.63	70.89	49.31	69.05/91.27	47.59/75.05	43.12/67.34

5 Results

5.1 Experimental Setup

We evaluate one frozen AWM state interface in three task spaces. The physical task uses OCP AUROC, balanced accuracy, and accuracy; the event task uses object, geometry, contact, event-time, and question-level metrics; the action task reports verb, noun, and joint action Top-1/Top-5 together with macro-F1 and seen-noun coverage. Dataset names, annotation details, split construction, and implementation settings are given in Appendix B; they are evaluation protocols rather than separate architectural components.

All direct readout comparisons use matched feature normalization, comparable parameter counts, identical optimization budgets, and the same split within a task space. The encoder, predictor, and HASP state extractor are frozen. Factor probes test information availability, task readouts test predictive sufficiency, and interventions test functional dependence. Input interventions modify visual context and rerun the frozen predictor; state interventions operate after HASP on slots, slot-time states, or pair instances. Target and non-target controls are reported whenever possible, since a global loss of information is not evidence of selective functional use. The experiments therefore separate four questions that are often conflated in representation evaluation: whether information is present, whether it is organized at the intended unit, whether it supports prediction, and whether the prediction selectively depends on it.

5.2 Main Results

Table 1 reports the current headline comparison. We use a common method vocabulary: AWM denotes the proposed HASP state interface with independent task readouts, while V-JEPA 2 denotes the corresponding conventional backbone-only baseline. External baselines are included when the protocol is available; empty cells are intentionally left unfilled rather than estimated.

The strongest available physical, event, and action records are obtained with AWM, while the analysis below asks why these numbers occur and whether the relevant information is present, sufficient, and functionally used. The V-JEPA2 row denotes the conventional backbone-only baseline, whereas the AWM row uses the proposed structured state interface. For CLEVRER, AWM improves option accuracy from 65.46 to 70.89 and question accuracy from 42.20 to 49.31 under the same validation protocol. ORCA is included as a reference because its physical protocol uses future-video latent evidence and is not strictly matched to current-only AWM. VideoMAE v2 reports 67.61 OCP AUROC, 62.60 balanced accuracy, and 62.86 accuracy on Physion++, together with the listed EK100 verb, noun, and action results. Both the ORCA and VideoMAE v2 physical references use ground-truth future-video latent evidence, so they are reported as contextual baselines rather than strictly causal comparisons to AWM’s context-only inference path.

Table 2: Native-factor readability at the intended state granularity.

State	Factor 1		Factor 2	
	Target	Score	Target	Score
Entity	Presence / AUROC	0.9634	Extent / R^2	0.5316
Dynamic	Speed / R^2	0.7198	Velocity / R^2	0.3805
Relation	Contact / AUROC	0.9789	Distance / R^2	0.5815
	Time-to-contact / R^2	0.5542		

5.3 Analysis: Testing the Abductive State Interface

The analysis is organized as one evidence chain rather than a collection of benchmark diagnostics. We first establish that the proposed states expose measurable factors, then test whether they support different kinds of prediction, next intervene on the evidence that should matter to the readout, and finally examine whether the same state interface can be reused when the task changes. Unless stated otherwise, probes and readouts have matched capacity and operate on frozen states.

5.3.1 Evidence is present at the proposed state levels

The first step is to test whether the proposed state factors contain the evidence their names imply. This is deliberately a representational test, not yet a claim about task use. We therefore train independent probes at the native granularity of each state: Entity slots for object factors, Dynamic slot–time states for motion factors, and Relation pair–time states for interaction factors. A positive result means that the evidence is available at the designated unit rather than only in a scene-level pooled shortcut.

Table 2 shows a coherent factor-to-unit pattern. Entity presence reaches 0.9634 AUROC and extent reaches $R^2 = 0.5316$, so the slots retain more than binary occupancy: they preserve coarse spatial geometry as well. For Dynamic, speed is more predictable ($R^2 = 0.7198$) than signed velocity ($R^2 = 0.3805$). This asymmetry is informative: motion magnitude is easier to recover than coordinate-dependent direction, suggesting that the state retains a stable temporal signal without implying a complete trajectory representation. Relation contact reaches 0.9789 AUROC, while pair distance and time-to-contact are also continuously decodable. Relation is therefore not merely a binary contact classifier; it exposes geometric and temporal interaction evidence. Taken together, the results establish information availability at the designated state units. They do not establish exclusivity, causal identification, or perfect temporal identity; the residual paths intentionally allow complementary information to remain readable across levels.

Additional slot diagnostics are retained for the appendix. They show high presence, target-object, and object-type accuracy, but imperfect identity consistency; slots should therefore be interpreted as an evidence-organizing inventory rather than guaranteed persistent trackers.

5.3.2 The state interface supports different kinds of prediction

Readability alone is insufficient: a state should also support consequences that are not explicitly used as probe targets. We therefore compare context-only and predicted-future evidence under matched-capacity readouts. The comparison asks not whether future evidence improves every metric, but which kinds of variables require a future hypothesis and which are already recoverable from the current latent.

Table 3. Effect of predicted-future evidence on present-state decoding.

Target	Context	Context + Future
Presence F1	99.58	99.77
Target accuracy	94.02	95.79
Object type accuracy	91.18	92.86

Table 4. Predicted-future evidence mainly improves interaction-time reasoning.

Metric	Context	Context + Future
Contact AUROC	96.2	97.2
Contact AP	20.4	—
First-contact accuracy	94.0	94.5
Collision-pair first-contact	17.1	25.4

Table 5: Probe-level object intervention selectively degrades the OCP readout.

Condition	Accuracy	Bal. Acc.	AUROC
Visual-only	0.6613	0.6595	0.7514
Visual + ShallowProbe	0.7175	0.7182	0.7962
Target mask	0.6325	0.6437	0.7391
Irrelevant mask	0.7088	0.7079	0.7834
Random slot	0.7163	0.7172	0.7930

The Physion++ comparison in Table 3 provides an important negative result. Context alone already yields 99.58 Presence F1, and predicted future raises it only to 99.77. Target-object and object-type accuracy improve by 1.77 and 1.68 points, respectively, but remain dominated by the current evidence stream. Thus predicted future is not a generic substitute for visual recognition: present-object semantics are largely available before the transition hypothesis is introduced. This result also prevents us from overstating the role of the predictor based on small aggregate gains.

The CLEVRER results in Table 4 reveal the complementary role of the predictor. Context-only decoding already obtains 96.2 contact AUROC, but collision-pair first-contact accuracy is only 17.1. Adding predicted future raises contact AUROC to 97.2 and collision-pair first-contact accuracy to 25.4, an 8.3-point gain on the difficult positive subset. The aggregate first-contact score changes much less because no-contact accuracy is 99.6. Moreover, contact average precision is only 20.4 despite the high AUROC, showing that ranking contact candidates is substantially easier than localizing the rare positive event. The predictor is therefore most useful for temporal disambiguation—whether an interaction will occur is already visible, while when it will occur benefits from a future hypothesis.

5.3.3 Intervention distinguishes information presence from information use

The preceding analyses establish availability and predictive value, but neither result shows that a task readout uses the evidence attributed to a particular state. We therefore intervene on evidence rather than on Attributor parameters. The protocols separate two causal distances: input interventions modify the visual context and rerun the frozen predictor and HASP, while probe-state interventions modify slots or pairs after HASP with the original visual latent fixed. Every intervention is paired with a random or non-target control, so a global loss of information is not mistaken for selective use.

Object-guided probe masks and temporal-static replacements test these two surfaces. The Physion++ ShallowProbe experiment modifies target-object tokens, trajectories and related pair fields while holding raw visual features fixed. Temporal staticification instead changes the input context and reruns the frozen predictor before recomputing HASP.

Table 5 first establishes that the structured probe adds information beyond the visual route: AUROC rises from 0.7514 for Visual-only to 0.7962 with the full ShallowProbe, and accuracy rises from 0.6613 to 0.7175. When the target object is masked, AUROC falls to 0.7391

Table 6: Temporal-static input intervention selectively degrades dynamic readouts.

Condition	Speed MAE	Contact AUROC	OCP AUROC
Baseline	0.0670	0.9837	0.7907
Temporal static	0.1120	0.9485	0.5646

Table 7: Relation-specific intervention affects contact evidence more than a matched control.

Condition	Contact AUROC	TTC MAE
Baseline	0.9633	0.1306
Contact-pair mask	0.9488	0.1376
Non-contact-pair mask	0.9518	0.1349

and accuracy to 0.6325. The corresponding irrelevant-object condition retains 0.7834 AUROC and 0.7088 accuracy, while random-slot injection leaves AUROC almost unchanged at 0.7930. The relevant comparison is therefore the target–non-target gap, not the absolute drop: it shows that the readout is sensitive to the object evidence attributed by the probe rather than to arbitrary token perturbations. Table 6 provides the temporal counterpart. Staticifying the input increases speed MAE from 0.0670 to 0.1120, lowers contact AUROC from 0.9837 to 0.9485, and reduces OCP AUROC to 0.5646. The asymmetric effects of spatial and temporal interventions are consistent with the proposed division of labor between Entity and Dynamic states. After context intervention, predictor regeneration preserves the same ordering: regenerated OCP AUROC is 0.7907 for the baseline, 0.7295 after object masking, and 0.6392 after temporal-static replacement. This supports propagation through the context $\rightarrow$ predictor $\rightarrow$ future $\rightarrow$ HASP path, although it is not a pixel-level causal effect because the masks also alter correlated patches and scene context.

At the state level, Relation provides a more local intervention surface. As shown in Table 7, masking a true contact pair lowers contact AUROC from 0.9633 to 0.9488 and increases TTC MAE from 0.1306 to 0.1376. A matched non-contact mask changes the same metrics to 0.9518 and 0.1349. The larger target-specific degradation supports functional selectivity, while the non-target effect indicates that residual and shared context prevent interpreting Relation as a causally isolated module.

The same logic applies to event queries, where the readout should use the object referred to by the question. For 3,557 predictive questions (7,114 answer options), we mask the relevant or an irrelevant object in the structured state while holding the original video and latent features fixed. In Table 8, relevant-object masking reduces question accuracy from 50.18 to 43.46 and option accuracy from 71.68 to 66.80, whereas irrelevant-object masking gives 47.34 and 69.55. The 3.26-point question-level and 2.76-point option-level gaps are evidence that the query readout follows object relevance encoded in Entity. The residual degradation under the irrelevant mask is also informative: the state retains shared scene and pair context, so Entity should be interpreted as functionally selective, not perfectly isolated. As above, this remains a probe-output intervention rather than a full visual counterfactual.

5.3.4 State factors are reusable across task spaces

The final test moves from functional dependence to reuse. We freeze the same state extractor and train equal-capacity readouts for a task whose labels are not native physical factors. This is a stronger test than transferring a classifier: the readout changes, but the Entity–Dynamic–Relation interface does not.

Table 9 shows a structured transfer pattern. Dynamic is the strongest single level for verb (56.98) and noun (33.03), while Entity remains competitive (50.43 verb and 31.61 noun). This asymmetry is consistent with the structure of the labels: verb is tied to action dynamics, whereas noun relies more on object identity and appearance. Relation is weaker in isolation (30.99 verb and 16.73 noun), not because it lacks information, but because EK100 labels

Table 8: Question-relevant object intervention in predictive event queries.

Condition	Question correct	Question acc.	Option correct	Option acc.
Baseline	1,785/3,557	50.18%	5,099/7,114	71.68%
Relevant-object mask	1,546/3,557	43.46%	4,752/7,114	66.80%
Irrelevant-object mask	1,684/3,557	47.34%	4,948/7,114	69.55%

Table 9: Cross-task reuse of individual state factors on EK100.

State level	Verb Top-1	Noun Top-1
Entity	50.43%	31.61%
Dynamic	56.98%	33.03%
Relation	30.99%	16.73%

are not primarily pairwise interaction labels. Conversely, Relation is the most natural state for physical contact and event-time reasoning. The results therefore support factor-selective reuse: a common interface can be read differently by different tasks, without requiring every task to use every state equally. They do not by themselves prove semantic disentanglement.

5.3.5 What the evidence establishes

Taken together, the analyses support AWM as a functionally organized state interface. Native probes show that Entity, Dynamic, and Relation expose measurable object, motion, and interaction factors at their designated units. The prediction studies show that current evidence is sufficient for much of the static semantics, while predicted future evidence contributes most to event-time localization. Intervention experiments then distinguish availability from use: target masks produce larger, factor-specific changes than non-target controls, both at the input pathway and in the structured state. Finally, the layer-wise EK100 results show that the same state extractor can be reused with task-specific combinations of factors.

These observations establish information availability, predictive utility, selective functional dependence, and cross-task reuse. They do not establish that the factors are mutually exclusive, that slots are perfect persistent tracks, or that the states are uniquely identifiable causal variables. Nor do they claim explicit posterior hypothesis generation; those are natural extensions of the present abductive interface.

5.4 The Abductive Loop: From Evidence to Consequence

The experiments can be read as successive views of one abductive loop. The model starts from partial visual evidence, attributes it to entity-, transition-, and relation-level states, and exposes the resulting interface to a consequence readout. We then intervene on the evidence that is supposed to support that consequence and ask whether the predicted change follows the same factor organization. Probes test whether an explanation is readable; task readouts test whether it is sufficient; and interventions test whether it is functionally used. Cross-task readouts finally test whether the interface survives a change in label space. Under this interpretation, AWM is not defined by one decoder score, but by the joint behavior of evidence, state, and consequence. The loop supports a functional abductive interpretation of HASP, while causal identification and explicit posterior hypothesis generation remain open.

6 Discussion and Limitations

The current evidence supports a scoped interpretation of AWM. Entity, Dynamic, and Relation contain measurable evidence at their intended granularities; the combined interface supports downstream prediction; and targeted input/state changes propagate selectively through the prediction path. These are evidence for functional organization, not a claim that the representation has recovered a unique causal factorization. At the same time, several limitations are important. Residual connections mean that factors are complementary rather than strictly orthogonal. Slot identity is not perfectly persistent, noun labels are long-tailed, and controlled interventions may alter correlated scene evidence. The frozen-probe protocol validates the abductive criterion retrospectively and does not identify true causal variables. A natural next step is a warm-start conditional flow or multi-hypothesis transition module that turns the conceptual score in Eq. 3 into explicit candidate generation and calibrated ranking. Held-out compositions and environments are also needed to test whether the interface reuses factors rather than memorizing co-occurrences.

7 Conclusion

AWM reframes a visual world model as an analyzable state interface. Rather than equating a world state with a single predictive latent, it asks whether visual evidence is organized into factors that can be decoded, used for prediction, and changed consistently under targeted intervention. HASP implements this idea with a conditional Entity–Dynamic–Relation hierarchy on top of a frozen predictive video backbone. The current results show that the same interface can expose object, motion, and interaction evidence and feed distinct physical, event, and action readouts. The broader contribution is an evaluation principle: world-state quality should be established by a joint explainability–predictability–intervention protocol, with explicit boundaries on causal and abductive claims.

AI Use Statement

Generative AI tools were used to assist with literature organization and editorial drafting. All technical claims, equations, experimental numbers, and code references were checked against the project files by the authors, who take responsibility for the final manuscript.

Reproducibility Statement

The source code, dataset-isolated training protocols, analysis scripts, and experiment notes are maintained in the accompanying `recipe_formal` directory. The analysis scripts export reusable state features and reproduce the factor probes and intervention tables without modifying training checkpoints.

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A Inference Procedure

For completeness, the deterministic forward computation used by all task readouts is listed below. This appendix gives the implementation-level ordering for the evidence-routing computation described in the Method section.

$$Z_c = E(X_{1:t}), \quad \hat{Z}_f = P(Z_c), \quad (25)$$

$$S^E = F_E(Z_c), \quad S^D = F_D(S^E, Z_c, \hat{Z}_f) + G_D(S^E, Z_c), \quad (26)$$

$$S^R = F_R(S^E, S^D, Z_c, \hat{Z}_f) + G_R(S^D, Z_c), \quad \hat{Y}_\tau = R_\tau(S^E, S^D, S^R). \quad (27)$$

The encoder, predictor, and state extractor are frozen in the analysis protocol; only the task-specific readout R_τ is trained. Input interventions rerun the left-hand side from the modified context, whereas state interventions act after S_t has been constructed.

Table 10: Additional diagnostics of the Physion++ Entity slots.

Diagnostic	Value
Presence F1	99.77
Slot occupancy	64.52
Identity consistency	70.29
ID switch rate	4.05
Target-object accuracy	95.79
Object-type accuracy	92.86
Color RGB MAE	0.2244
Center MAE	0.0454
Geometry MAE	0.0285
Velocity MAE	0.00629

Table 11: Physion++ OCP confusion counts under probe interventions.

Condition	TP	TN	FP	FN
Visual-only	237	292	129	142
Visual + ShallowProbe	277	297	124	102
Target mask	325	181	240	54
Irrelevant mask	262	305	116	117
Random slot	279	294	127	100

B Additional Experimental Details

The physical experiments use the Physion++ annotations and evaluate OCP prediction with the established train/validation/test protocol. The event experiments use CLEVRER validation windows with context starts at 0, 32, and 64; current-only settings never read real future frames or future latent features. The action experiments use the official EPIC-KITCHENS-100 training annotations, split by video into 90% probe training and 10% probe validation, and reserve the official validation annotations as final test data. Noun classes unseen in the training vocabulary are excluded from closed-set noun accuracy and reported as unknown-noun coverage.

All task heads are trained independently with AdamW. Unless otherwise noted, the encoder, predictor, and HASP state extractor are frozen, and only the task readout is optimized. For action recognition, the model reports separate verb and noun probabilities and a closed-set joint action score formed from their logits. Complete class counts, cache specifications, intervention masks, and command lines are provided with the reproducibility package.

B.1 Additional Physion++ diagnostics

The factor probes expose strong object-level semantics, but slot identity is not perfectly persistent across time. Table 10 reports the diagnostic set used to qualify the Entity interpretation. Intervention confusion counts are given in Table 11. Target masking increases positive recall while sharply reducing negative recall, indicating a decision-boundary shift rather than simple disappearance of positive evidence. At the paired-logit level, Table 12 shows that target masking changes the absolute logit about 3.9 times more than irrelevant masking and 7.6 times more than random-slot injection.

B.2 Baseline diagnostics and protocol gaps

The VideoMAE v2 Physion++ reference reaches 65.68 F1 on the test split (TP/TN/FP/FN = 423/325/241/201), but uses ground-truth future latent features and is therefore an oracle-future representation baseline. The corresponding ORCA reference has F1 62.93 with TP/TN/FP/FN = 438/236/330/186. On EK100, the direct-head VideoMAE v2 baseline has a large train-validation gap, summarized in Table 13; this is consistent with overfitting under the single-clip protocol.

Table 12: Paired OCP-logit changes under probe-level interventions.

Condition	Mean change	Mean absolute change
Target mask	+0.49195	0.67680
Irrelevant mask	-0.08051	0.17355
Random slot	+0.02437	0.08895

Table 13: VideoMAE v2 EK100 direct-head baseline: train-validation gap.

Split	Verb Top-1	Noun Top-1	Action Top-1
Train	96.70	96.68	92.92
Probe validation	59.54	46.37	35.96

For CLEVRER, the high contact AUROC should be interpreted together with its 20.4 contact average precision and 99.6 no-contact accuracy. These imbalance diagnostics motivate reporting collision-pair first-contact accuracy separately in the main text.